

# Sourav Ray

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## Professional Summary

Highly accomplished Data & AI Engineer with 3+ years of experience and M.Tech degree in Operations Research, Sourav Ray excels at designing production data engineering pipelines, ensuring pipeline reliability through robust testing and validation. Leveraging expertise in Python, PySpark, SQL, Airflow, Snowflake, AWS, and AI/ML frameworks like OpenAI, Gemini, and LangChain, Sourav delivers scalable solutions for enterprise customers, driving 60% increase in data accessibility.

## Highlights

- Promoted within 2 years for consistently exceeding performance expectations, driving high-impact Data Engineering and GenAI/Agentic AI initiatives.
- 3+ years of experience building Snowflake-based Data Mesh, Governance, and GenAI solutions, spanning both core data platform engineering and applied AI/ML delivery.
- Built multi-agent, MCP-based AI orchestration systems integrating LLMs with enterprise platforms (ServiceNow, Snowflake, observability tools) for automated investigation and root-cause analysis.
- Hands-on experience with LangChain, RAG, vector search, and LLM-based inference for schema mapping, semantic search, and automated RCA.
- SnowPro Advanced Architect, SnowPro Advanced Data Scientist, and AWS Solutions Architect Associate certified.
- Proven track record delivering enterprise-scale data quality, governance, and automation platforms; mentored junior engineers on DBT development, data quality frameworks, and GenAI-based automation best practices.

## Technical Skills

**Cloud & Data Platforms:** AWS (S3, Lambda, Kinesis, Firehose, DMS), Snowflake (Snowpark, Snowpipe), GCP (Vertex AI, BigQuery)

**Data Engineering:** DBT, Apache Airflow, Coalesce, SQL, PySpark, Kafka, ETL/ELT, Data Contracts, Data Mesh

**AI / ML & GenAI:** OpenAI, Gemini, LangChain, RAG, ChromaDB, NLP, TensorFlow, Time-Series Forecasting, Genetic Algorithms, Multi-Agent Orchestration, Model Context Protocol (MCP)

**Programming:** Python, SQL, Shell Scripting

**Tools:** Docker, Kubernetes, Git, GitHub Actions, Sigma

## Professional Experience

**Factspan Analytics, Bengaluru**

*Data & AI Engineer*

Sep 2023 – Present

**Customer 360 Data Mesh & Governance | Snowflake, AWS, DBT**

- Architected a containerized Python Data Contract framework supporting RDBMS, flat-file, and streaming ecosystems, improving enterprise data accessibility by 60%.
- Engineered metadata-driven automation generating Silver and Gold DBT artifacts (.sql, .yaml, .md), reducing development effort by 80%.
- Designed ingestion frameworks using AWS DMS, Kinesis, Firehose, S3, and Snowpipe for near real-time onboarding.
- Built multi-stage reconciliation pipelines using row-count validation and MD5-based record verification to ensure data fidelity.
- Developed automated RCA tooling identifying column-level discrepancies and format anomalies.
- Validated SAP BusinessObjects reports against Snowflake models ensuring reporting consistency across legacy and modern platforms.
- Governed finance datasets including GL, revenue, and transactional domains through audit-ready validation controls.

**AI-Powered Network Incident Orchestrator | Python, MCP, LLM Agents**

- Built a multi-agent orchestrator that coordinates specialized AI agents to accelerate network incident investigation and root cause analysis (RCA) across ServiceNow, Infoblox, NetBox, SevOne, Datadog, Splunk, and LogicMonitor.
- Designed per-platform agents, each backed by its own Model Context Protocol (MCP) server, providing a consistent abstraction layer over heterogeneous observability tools and device-management APIs, including secure SSH-based access to Cisco and non-Cisco network devices.
- Implemented dynamic agent selection and parallel invocation logic that correlates multi-source data – incidents, logs, metrics, topology, device health, and configuration – into a single consolidated RCA report.
- Improved performance and scalability through in-memory caching and per-agent concurrency controls (capped at 3 concurrent users), reducing Mean Time to Detect (MTTD) and Mean Time to Resolve (MTTR).

**GenAI Schema Validator & Automated Mapping Platform**

- Developing an enterprise-scale schema and report matching platform using Python, Vector Search, and LLMs, implementing Exact Match, Semantic Match, and LLM-based inference layers for automated column mapping.
- Built governance controls to identify and exclude masked PII attributes, and leveraged LLMs to auto-generate metadata descriptions for undocumented fields.

- Leading UAT and validation across complex enterprise datasets and reporting structures.

#### **Data-Powered Assist Chatbot**

- Developed Gold-layer models supporting Sigma lineage analytics.
- Enabled semantic querying in Snowflake through NLP-powered chatbot workflows.
- Built ETL pipelines using Coalesce and integrated image URI-based workflows.

#### **Education**

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**NIT Durgapur** | M.Tech, Operations Research | CGPA: 8.36/10 | 2021–2023

**GCETTB** | B.Tech, Textile Technology | CGPA: 7.43/10 | 2015–2019

#### **Certifications**

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- Snowflake SnowPro Advanced Architect
- Snowflake SnowPro Advanced Data Scientist
- AWS Certified Machine Learning – Specialty
- AWS Certified Solutions Architect – Associate